

Write this expression as a sum of two integrals, each of which you can integrate.

It is just the above integral with different limits:

$$\int_0^1 x^2 dx = \frac{1}{3}x^3 \Big|_0^1 = \frac{1}{3} \quad (10.1)$$

$$\int_1^2 x^2 dx = \frac{1}{3}x^3 \Big|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3} \quad (10.2)$$

$$\int_0^2 x^2 dx = \int_0^1 x^2 dx + \int_1^2 x^2 dx = \frac{1}{3} + \frac{7}{3} = \frac{8}{3} \quad (10.3)$$

where $\frac{8}{3}$ is the value of the integral. The above calculation is correct, even though the two integrals given for the second one do not integrate to $\frac{1}{3}x^3$.

It is not correct to take the upper limit as zero. The $\frac{1}{3}x^3$ formula for the integral is only valid for $x \geq 0$, and the integral is only valid for $x \geq 0$. The integral is only valid for $x \geq 0$.

$$\begin{aligned} \int_0^1 x^2 dx &= \frac{1}{3}x^3 \Big|_0^1 = \frac{1}{3} \\ \int_1^2 x^2 dx &= \frac{1}{3}x^3 \Big|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3} \\ \int_0^2 x^2 dx &= \frac{1}{3}x^3 \Big|_0^2 = \frac{8}{3} \end{aligned} \quad (10.4)$$

Then the value of the integral is $\frac{8}{3}$, which is the same as the value of the integral from 0 to 2. The integral is only valid for $x \geq 0$.

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$$\int_0^1 x^2 dx = \frac{1}{3}x^3 \Big|_0^1 = \frac{1}{3} \quad (10.5)$$

The value of the integral is $\frac{8}{3}$, which is the same as the value of the integral from 0 to 2. The integral is only valid for $x \geq 0$.

10.4 The value of the integral is the same as the value of the integral

Because the function $f(x) = x^2$ is continuous on the interval $[0, 2]$, the value of the integral is the same as the value of the integral from 0 to 2.

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$$\int_0^1 x^2 dx = \frac{1}{3}x^3 \Big|_0^1 = \frac{1}{3}$$

The value of the integral is $\frac{8}{3}$, which is the same as the value of the integral from 0 to 2.

$$\int_1^2 x^2 dx = \frac{1}{3}x^3 \Big|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$$

Keywords: *depression, mood, mood disorder, mood disorder diagnosis, mood disorder treatment, mood disorder symptoms, mood disorder risk factors, mood disorder prevalence, mood disorder incidence, mood disorder etiology, mood disorder pathophysiology, mood disorder epidemiology, mood disorder clinical presentation, mood disorder differential diagnosis, mood disorder management, mood disorder prognosis, mood disorder research, mood disorder clinical practice, mood disorder patient care, mood disorder health care, mood disorder public health, mood disorder social issues, mood disorder cultural issues, mood disorder legal issues, mood disorder ethical issues, mood disorder policy issues, mood disorder education, mood disorder training, mood disorder continuing education, mood disorder professional development, mood disorder research funding, mood disorder research methods, mood disorder research findings, mood disorder research implications, mood disorder research future directions.*

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She is a... *[illegible]*

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